Section 5

Thursday, February 12, 2026 4:37 PM

- Connection Between Binomial and Poisson Distribution

$X \sim Bin(n, p)$ $x = 0, 1, \ldots, n$ $Y \sim Pois(\lambda)$ $y \geq 0$

# of success out of n trials — # trials — probability of success

expected # events/ successes in a given time period

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \qquad P(Y=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

★ Poisson Approximation to Binomial

lecture showed that:

letting $\lambda = np \rightarrow p = \frac{\lambda}{n}$

$$\lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k} = \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \frac{\lambda^k e^{-\lambda}}{k!}$$

n to be large ≥ 100
p to be small
$\Big\}$ $n \cdot p = \lambda \leq 10$

Exercises: Take 10 standard decks of cards, randomly select 200 → 520 cards total

1) what is the probability you have 2 Ace of Hearts?

a) determine if it's appropriate to use the Poisson approximation

b) Write the probability in terms of the Binomial pmf

$p = \frac{1}{52}$, $n = 200$

X = # Ace of Hearts

$$P(X=2) = \binom{100}{2} \left(\frac{1}{52}\right)^2 \left(\frac{51}{52}\right)^{98} \approx$$

c) Write the Probability in terms of the Poisson

$\lambda = n \cdot p = \frac{200}{52}$

$$P(X=2) = \frac{\left(\frac{200}{52}\right)^2 e^{\frac{200}{52}}}{2!} \approx$$

2) What is the probability you have 1 Red King

a)

b) $n = 200$ $p = \frac{1}{26}$